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Centers for Medicare and Medicaid Services
U.S. Department of Health and Human Services
7500 Security Boulevard
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RE: CMS-0041-NC, Request for Information; Health Technology Ecosystem

Podimetrics thanks the Centers for Medicare and Medicaid Services (CMS) for the opportunity to submit comments on the "Request for Information; Health Technology Ecosystem (CMS-0041-NC).

Based in Somerville, Massachusetts, Podimetrics is a tech-enabled services company dedicated to preventing diabetic foot ulcers (DFUs) and associated amputations, which are among the most costly and debilitating complications of diabetes. The company's innovative SmartMat™ digital health technology uses evidence-based remote temperature monitoring (RTM)[1] for early detection and prevention of DFUs and associated amputations. Data from Podimetrics' partnership with the Veterans Health Administration – The Initiative to End Diabetic Limb Loss at the VA (TIEDLVV)[2 3 4] – as well as with commercial health plans[5] shows that use of the SmartMat™ leads to significant reductions in potentially avoidable and costly hospitalizations, substantial declines in total healthcare spending, and meaningful improvements in patient mortality.[6 7]

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[1] Dy SM et al. Preventing Complications and Treating Symptoms of Diabetic Peripheral Neuropathy. Comparative Effectiveness Review No. 187. (Prepared by the Johns Hopkins University Evidence-based Practice Center under Contract No. 290-2015-00006-I.) AHRQ Publication No. 17-EHC005-EF. Rockville, MD: Agency for Healthcare Research and Quality; March 2017.
[2] VA News. How Innovation and Partnership are Ending Diabetic Limb Loss at VA. May 21, 2020.
[3] Lutes, T. VA's remote temperature monitoring prevents diabetic limb loss. VA News. December 19, 2023.
[4] Bloom, J. Testimony before the United States House of Representatives. Committee on Veterans' Affairs Hearing on Harnessing Biomedical Innovation: Modernizing VA Healthcare for the Future. April 1, 2025.
[5] Littman, A. J., Lindsley, K. B., Boyko, E. J., Sangeorzan, B. J., Reiber, G. E., Garrett, N., Skalsky, A. J., & Raugi, G. J. Evaluation of the Effectiveness of Remote Foot Temperature Monitoring for Prevention of Amputation in a Large Integrated Health Care System. *Diabetes Care*, vol. 46, no. 8, 2023. Available at:https://doi.org/10.2337/dc22-1492.
[6] Frykberg, R. G., et al. "Feasibility and Efficacy of a Smart Mat Technology to Predict Development of Diabetic Plantar Ulcers." *Diabetes Care,* vol. 40, no. 7, 2017, pp. 973-980.
[7] Isaac et al. Lower resource utilization for patients with healed diabetic foot ulcers during participation in prevention program with foot temperature monitoring. *BMJ Open Diab Research & Care*. 2020.

Podimetrics strongly supports the work of CMS and the U.S. Department of Health and Human Services (HHS) Assistant Secretary for Technology Policy / Office of the National Coordinator for Health Information Technology (ASTP/ONC) to foster widespread adoption and utilization of digital health technology in federal health programs and the U.S. healthcare system broadly. Our experience with the VHA and commercial payers demonstrates the value and potential of innovative digital health technology to facilitate patient and provider engagement and prevent the onset of expensive and debilitating conditions like DFUs and diabetes-related amputations. To that end, we offer the following comments on advancing digital health technology in the healthcare system:

I. **Example of the high-value digital health technology:** Evidence-based SmartMat™ RTM digital health technology significantly reduces rates of costly and deadly diabetic foot ulcers and associated amputations.

II. **Policy recommendations to expand access to innovative digital health technology in Medicare and Medicaid**: CMS should: (1) establish a separate Medicare pathway for coverage of innovative digital health technology, including coverage for innovative RTM technology for at-risk diabetes patients and the work of healthcare providers in effectively treating and managing the care of patients benefitting from access to these novel technologies; and (2) implement quality measures in Medicare and Medicaid to foster use of high-value digital health technology that empowers patients to engage in their healthcare and improve outcomes.

Podimetrics' comments reflect our strong desire and commitment to work with CMS and the ASTP/ONC to facilitate widespread access and adoption of digital health technology in order to foster patient and provider engagement, improve health outcomes, and lower total healthcare spending.

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I. Example of High-Value Digital Health Technology: Evidence-Based SmartMat™ Prevents Diabetic Foot Ulcers and Associated Amputations

Background: According to the American Diabetes Association (ADA), 160,000 diabetes-related amputations occur annually[8] – a 75 percent increase over the past decade[9] – and up to 85 percent of those amputations are preventable but occur due to patient challenges accessing high-quality care.[10] Diabetic foot ulcers and diabetes-related amputation are among the most costly and debilitating complications of the disease:

- **$3.5 billion in single-year Medicare spending on potentially avoidable hospitalizations related to diabetes amputations:** In 2020 alone, Medicare spent $3.5 billion on potentially avoidable hospitalization costs from diabetes-related amputations.[11] In that same year, Medicare spent another nearly $1 billion on potentially avoidable hospitalization costs for beneficiaries with diabetes experiencing both amputations and ketoacidosis[12] – suggesting that, if those costs continue, ***Medicare potentially could spend upwards of $40 billion or more over 10 years on potentially unnecessary and avoidable care for diabetes-related amputations***. Indeed, the cost of a single amputation associated with a diabetic foot ulcer can be more than $100,000 and the mean one-year cost from a public payor perspective for a diabetic foot ulcer is $44,200 (2015 USD).[13] [14]

- **High risk of mortality for patients with diabetes-related amputation:** Patients suffering from diabetes-related amputation have higher five-year mortality rates than patients with coronary artery disease, breast cancer, and colorectal cancer.[15] They also are much more

[8] *Ibid*.
[9] ADA. American Diabetes Association Shares Legislative Priorities with the 119th Congress. January 24, 2025.
[10] ADA. Press Release: American Diabetes Association Unveils Amputation Prevention Alliance to Address Diabetic-Related Amputation Pandemic. September 22, 2022.
[11] 88 FR 52262
[12] 88 FR 52262
[13] Gorman, Anna. Diabetic Amputations A 'Shameful Metric' Of Inadequate Care. California Healthline. May 1, 2019.
[14] Chan et al. Cost-of-illness studies in chronic ulcers: a systematic review. *J Wound Care.* 2017 Apr 1;26(sup4):S4-S14. Doi: 10.12968/jowc.2017.26.Sup4.S4.
[15] ADA. Amputation Prevention Alliance. Accessed April 23, 2025.

likely to have moderate to severe dysfunction in activities of daily living, such as walking, and are at increased risk for re-amputation and re-ulceration.[16]

- **Costly inpatient admissions for patients with diabetic foot ulcers:** Similar to diabetes-related amputations, diabetic foot ulceration often leads to potentially costly and avoidable hospitalization and patient mortality. In a study of nearly 80,000 Medicare beneficiaries who had at least one episode of diabetic foot ulceration, all-cause mortality was 1.5 times more likely and all-cause inpatient admission was 2.8 times more likely during the diabetic foot ulceration period compared to healing period afterwards in the same beneficiaries. Diabetic foot ulceration increased the likelihood of inpatient admission for cardiovascular, renal, and pulmonary complications and potentially triggered a number of excess acute-on-chronic complications. Among the 30 most common admission MS-DRGs, all but one were more frequently observed during the diabetic foot ulceration episode than in the healing period following that episode.[17]

SmartMat™ RTM digital health technology prevents DFUs and amputations: Published data shows that use of remote temperature monitoring reduces the incidence of diabetic foot ulceration – the leading cause of amputation – by approximately 70 percent.[18] [19] [20] As a result, a ***comparative effectiveness review from the Agency for Healthcare Research and Quality (AHRQ)[21] and three (3) clinical practice guidelines[22] [23] [24] currently recommend use of remote temperature monitoring for diabetes patients at risk for amputation***.

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[16] Frykberg et al. Feasibility and Efficacy of Smart Mat Technology to Predict Development of Diabetic Plantar Ulcers. *Diabetes Care.* 2017 Jul;40(7):973-980. DOI: 10.2337/dc16-2294

[17] Petersen et al. Higher rates of all-cause mortality and resource utilization during episodes-of-care for diabetic foot ulceration. Diabetes Res Clin Pract. 2022 February; 184: 1091832. doi:10.1016/j.diabres.2021.109182.

[18] Lavery LA, Higgins KR, Lanctot DR, et al. Home monitoring of foot skin temperatures to prevent ulceration. Diabetes Care 2004;27:2642–7.

[19] Frykberg et al. Feasibility and Efficacy of Smart Mat Technology to Predict Development of Diabetic Plantar Ulcers. *Diabetes Care.* 2017 Jul;40(7):973-980. DOI: 10.2337/dc16-2294

[20] Lavery LA, Higgins KR, Lanctot DR, et al. Preventing diabetic foot ulcer recurrence in high-risk patients: use of temperature monitoring as a self-assessment tool. Diabetes Care 2007;30:14–20.

[21] Dy SM et al. Preventing Complications and Treating Symptoms of Diabetic Peripheral Neuropathy. Comparative Effectiveness Review No. 187. (Prepared by the Johns Hopkins University Evidence-based Practice Center under Contract No. 290-2015-00006-I.) AHRQ Publication No. 17-EHC005-EF. Rockville, MD: Agency for Healthcare Research and Quality; March 2017.

[22] Frykberg RG, Zgonis T, Armstrong DG, et al. Diabetic foot disorders. A clinical practice guideline (2006 revision). J Foot Ankle Surg 2006;45:S1–66.

[23] Lavery LA, Davis KE, Berriman SJ, et al. WHS guidelines update: diabetic foot ulcer treatment guidelines. Wound Repair Regen 2016;24:112–26.

[24] Bus SA, Lavery LA, Monteiro-Soares M, et al. Guidelines on the prevention of foot ulcers in persons with diabetes (IWGDF 2019 update). Diabetes Metab Res Rev 2020;36 Suppl 1:e3269.

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Podimetrics' innovative SmartMat™ technology employs remote temperature monitoring for early detection and prevention of DFUs and associated amputations. The patient stands on the SmartMat™ for 20 seconds per day from the comfort of home while the technology scans for temperature changes in the feet that could indicate a developing foot ulcer. The data then is transmitted securely via cellular network for analysis without the need for Wi-Fi or a mobile device, allowing a clinician to intervene in time to prevent the ulcer before it appears in most cases. In peer-reviewed research from Kaiser Permanente, benefits of the SmartMat™ included[25]:

- 71 percent reduction in **diabetic amputations**
- 52 percent reduction in **all-cause hospitalizations**
- 40 percent reduction in **emergency department (ED) visits**
- 26 percent reduction in **all-cause outpatient visits**

VA Program with SmartMat™ reduces DFUs and amputations for veterans: Podimetrics and the U.S. Department of Veterans Affairs entered into a formal partnership in 2020 to provide at-risk veterans with diabetes access to the SmartMat™ RTM technology.[26] [27] [28] [29] The Initiative to End Diabetic Limb Loss at the VA (TIEDLVV) has been highly successful:

- **$100 million in healthcare savings in first 12 months:** Data from 6,428 Veterans suggested the VA saved roughly $100 million over the first 12 months of enrollment from avoided hospitalizations and outpatient visits alone – translating to savings of nearly $16,000 per veteran per year.[30]

- **Mortality reduction:** Veterans had a 37 percent reduction in one-year mortality over the first 12 months of enrollment in the program – equating to one veteran life saved for every 23 veteran participants who entered the program.[31]

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[25] Isaac et al. Lower resource utilization for patients with healed diabetic foot ulcers during participation in prevention program with foot temperature monitoring. *BMJ Open Diab Research & Care*. 2020.
[26] VA Connected Care. VA's Remote Temperature Monitoring Program Prevents Diabetic Limb Loss. December 21, 2023.
[27] VA Diffusion Marketplace. Reducing Hospital Admissions and Amputative Prevention: Remote Temperature Monitoring. Accessed April 23, 2025.
[28] VA. VA Focuses on Preventive TelePodiatry Care. July 26, 2023.
[29] VA. SmartMat delivers preventive TelePodiatry Care. July 24, 2023.
[30] Bloom, J. Testimony before the United States House of Representatives. Committee on Veterans' Affairs Hearing on Harnessing Biomedical Innovation: Modernizing VA Healthcare for the Future. April 1, 2025.
[31] Littman, A. J., Lindsley, K. B., Boyko, E. J., Sangeorzan, B. J., Reiber, G. E., Garrett, N., Skalsky, A. J., & Raugi, G. J. Evaluation of the Effectiveness of Remote Foot Temperature Monitoring for Prevention of Amputation in a Large Integrated Health Care System. *Diabetes Care*, vol. 46, no. 8, 2023. Available at:https://doi.org/10.2337/dc22-1492.

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- **Early detection of DFUs:** Clinicians detected 97 percent of diabetic foot ulcers over five weeks before they otherwise would have identified them – resulting in numerous veterans avoiding debilitating and deadly DFUs and associated amputations.[32]

II. Policy Recommendations to Advance Digital Health Technology

Podimetrics strongly applauds the efforts of CMS and ASTP/ONC to facilitate widespread adoption and access to digital health technology in federal health programs and the U.S. healthcare system more broadly. To that end, Podimetrics respectfully offers the following policy recommendations:

1. **Medicare coverage pathway for innovative digital health technology:** In 2024, CMS established a limited pathway for Medicare coverage of a small number of technologies and devices called Transitional Coverage for Emerging Technologies (TCET).[33] TCET replaced the much more expansive medical device coverage pathway established in the first term of the Trump Administration, Medicare Coverage of Innovative Technology (MCIT).[34] CMS should build on the work of the first term of the Trump Administration to significantly expand TCET and establish a broad separate pathway for Medicare coverage of digital health technologies. Unlike MCIT, the pathway should include digital health technologies beyond only those with "Breakthrough" designation from the Food and Drug Administration (FDA) so that Medicare beneficiaries have access to truly innovative devices and technologies that improve health outcomes like the SmartMat™ RTM technology that prevents debilitating, costly and deadly DFU and amputation complications from diabetes. As part of this expansive pathway, Medicare should cover and recognize the work of healthcare providers to effectively treat and manage patients benefitting from these novel technologies.

2. **Medicare and Medicaid quality measures:** Two Medicare Merit-based Incentive Payment System (MIPS) quality measures address diabetes-related foot care, but do not assess prevention of DFUs and associated amputation. In Medicare Advantage (MA), the Health Effectiveness Data and Information Set (HEDIS) Comprehensive Diabetes Care measure does not evaluate foot care. Similarly, the 2023 and 2024 Core Set of Adult Health Care Quality Measures for Medicaid includes four diabetes-related measures, but none that address quality of care related to DFUs and amputations. Therefore, CMS should work with

[32] Frykberg, R. G., et al. "Feasibility and Efficacy of a Smart Mat Technology to Predict Development of Diabetic Plantar Ulcers." *Diabetes Care,* vol. 40, no. 7, 2017, pp. 973-980.
[33] CMS. Fact Sheet: Final Notice – Transitional Coverage for Emerging Technologies (CMS-3421-FN). August 7, 2024.
[34] 86 FR 2987

stakeholders to develop new quality measures for Medicare and Medicaid that specifically assess: (1) reducing rates of DFUs in at-risk patients with diabetes; and (2) reducing rates of DFU-related amputations in at-risk patients with diabetes.